LIBXS Samples

Batched GEMM

Exercises the LIBXS batched GEMM API (`libxs_gemm.h`) in several flavors: strided, pointer-array, indexed, and grouped. All programs are multithreaded via OpenMP, report wall-clock time and GFLOPS/s, and optionally validate results via `libxs_matdiff`.

Building

```
cd samples/gemm
make GNU=1
```

LIBXS must be built first from the repository root.

Programs

gemm_strided Strided batch DGEMM: all matrices packed contiguously with constant stride. Uses `libxs_gemm_index` with index_stride=0.

```
./gemm_strided.x [M [N [K [batchsize [nrepeat [beta [pad]]]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3, beta=1.0, pad=0.

gemm_batch Pointer-array batch DGEMM: matrices accessed through pointer arrays. Supports a duplicate C-matrix mode to exercise lock-forward sync.

```
./gemm_batch.x [M [N [K [batchsize [nrepeat [dup [beta [pad]]]]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3, dup=0, beta=1.0, pad=0.

dup	Mode	Description
0	None	Each C-matrix is unique (LIBXS_GEMM_FLAG_NOLOCK)
1	Sorted	Half-unique C-pointers, duplicates consecutive
2	Shuffled	Half-unique C-pointers, randomly shuffled

gemm_index Index-array batch DGEMM: matrices in contiguous buffers, addressed through explicit element-offset index arrays (ia, ib, ic).

```
./gemm_index.x [M [N [K [batchsize [nrepeat [beta [pad]]]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3, beta=1.0, pad=0.

gemm_indexf Fortran variant of gemm_index. Demonstrates the LIBXS Fortran module interface with one-based index arrays and C_LOC/C_SIZEOF interop. Requires a Fortran compiler.

```
./gemm_indexf.x [M [N [K [batchsize [nrepeat]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3.

gemm_groups Grouped batch DGEMM: multiple groups of different matrix shapes dispatched in sequence, each with its own `libxs_gemm_config_t`. Per-group dimensions grow by 4 starting from base_m.

```
./gemm_groups.x [ngroups [batch_per_group [nrepeat [base_m [beta [pad]]]]]]
```

Defaults: ngroups=2 (max 4), batch_per_group=30000, nrepeat=3, base_m=8, beta=1.0, pad=0.

Validation

Set the CHECK environment variable to validate results:

```
CHECK=1 ./gemm_strided.x
CHECK=1 ./gemm_batch.x 23 23 23 1000 1 2
CHECK=1 ./gemm_index.x
CHECK=1 ./gemm_groups.x
```

When padding is enabled (pad>0), the check also verifies that leading-dimension padding in C-matrices has not been overwritten.

Memory Comparison Benchmark

Benchmarks `libxs_diff`, `libxs_memcmp`, and the C standard library's `memcmp` for both large contiguous comparisons and many small fixed-size comparisons. A Fortran variant compares `libxs_diff` against Fortran array intrinsics.

Building

```
cd samples/memory
make GNU=1
```

Produces `memcmp.x` (C) and, if a Fortran compiler is found, `memcmpf.x` and `matcpyf.x`.

Usage (C)

```
./memcmp.x [elsize [stride [nelems [niters]]]]
```

Argument	Default	Description
elsize	INSIZE/MAXSIZE (compile, 64)	Element comparison size in bytes
stride	max(MAXSIZE, elsize)	Distance between successive elements
nelems	2 GB / stride	Number of elements
niters	5	Repetitions (arithmetic average)

Environment Variables

Variable	Default	Description
STRIDED	0	0: one-call when stride==elsize, else loop; 1: seq loop; >=2: random
CHECK	0	Non-zero: validation pass (inject diffs, cross-check implementations)
MISMATCH	0	0: equal buffers; 1: first byte; 2: middle; 3: last; >=4: random
OFFSET	0	Shift buffer b off its natural alignment by this many bytes

Compile-Time Knobs

Macro	Default	Description
MAXSIZE	64	Stride floor (elements spaced >= this far)
INSIZE	MAXSIZE	Default element size when argument is omitted

```
## 32-byte elements, sequential strided access, 10 repetitions
./memcmp.x 32 0 0 10

## 8-byte elements, random traversal, 3 repetitions
STRIDED=2 ./memcmp.x 8 0 0 3

## contiguous (single-call) comparison, ~2 GB
./memcmp.x 64 64 0 5

## mismatch at last byte, buffer b misaligned by 3 bytes
MISMATCH=3 OFFSET=3 ./memcmp.x 32 0 0 5
```

Environment variables and compile-time macros apply only to `memcmp.x`.

Usage (Fortran -- memcmpf)

```
./memcmpf.x [nelements [nrepeat]]
```

Element type is hard-coded as INTEGER(4). Compares ~2 GB by default. Reports per-iteration times (ms) and average throughput (MB/s).

Usage (Fortran -- matcpyf)

```
./matcpyf.x [m [n [ldi [ldo [nrepeat [nmb]]]]]]
```

Argument	Default	Description
m	4096	First matrix dimension
n	m	Second matrix dimension
ldi	m (enforced m)	Leading dimension of input
ldo	ldi	Leading dimension of output
nrepeat	2	Number of repetitions
nmb	2048	Memory budget in MB

Benchmarks `libxs_matcopy` and `libxs_matcopy_task` for matrix copy and zeroing. When the matrix is square and ldi==ldo, also benchmarks `libxs_otrans` and `libxs_otrans_task` for out-of-place transpose.

What Is Measured

Three comparison implementations are benchmarked:

- `libxs_diff` -- SIMD-dispatched short-buffer comparison (SSE/AVX2/AVX-512). Limited to elsize < 256 bytes, per-element.
- `libxs_memcmp` -- SIMD-dispatched memcmp replacement. Single call for contiguous buffers, or per-element for strided access.
- stdlib `memcmp` -- C library implementation under the same pattern.

Each kernel gets freshly initialized data before its timed section. An untimed warm-up pass runs before the first measured iteration. The summary reports min / median / max across iterations plus peak MB/s. Reported MB/s counts both buffers (2 * nbytes / time).

Ozaki-Scheme Low-Precision GEMM

Intercepts DGEMM, SGEMM, ZGEMM, and CGEMM at link time, replacing them with low-precision Ozaki schemes (mantissa slicing or CRT). Real GEMM calls go through the selected scheme; complex GEMM (ZGEMM/CGEMM) is mapped to real GEMM internally.

Build

```
cd samples/ozaki
make GNU=1 -j $(nproc)
```

LIBXS must be built first from the repository root. When a sibling `libxstream` directory is detected, the optional OpenCL/GPU path is compiled in (see OZAKI_OCL below).

Link-Time Interception

Two link-time variants are built per precision:

- `dgemm-blas.x` / `sgemm-blas.x` -- dynamically linked against BLAS
- `dgemm-wrap.x` / `sgemm-wrap.x` -- linked with `--wrap=` (GNU ld)
- `zgemm-wrap.x` / `cgemm-wrap.x` -- complex GEMM wrappers

Static (`libwrap.a`) and shared (`libwrap.so`) wrapper libraries are built by default. The shared library can intercept any application:

```
LD_PRELOAD=/path/to/libwrap.so ./application
```

The static library requires explicit `--wrap` flags:

```
-Wl,--wrap=dgemm_ -Wl,--wrap=sgemm_ -Wl,--wrap=zgemm_ -Wl,--wrap=cgemm_
```

Test scripts: `test-wrap.sh` exercises all variants, `test-check.sh` validates both schemes, `test-mhd.sh` tests MHD file-based input.

Test Driver

```
dgemm-wrap.x [A.mhd|M [B.mhd|N [K [TA [TB [ALPHA [BETA [LDA [LDB [LDC]]]]]]]]]]
```

Defaults: M=N=K=257, TA=TB=0, ALPHA=BETA=1.0. The first argument can be 0 followed by MHD filenames to load A and B matrices from files. The `zgemm-wrap.x` and `cgemm-wrap.x` drivers call ZGEMM and CGEMM.

Environment Variables

Scheme Selection

Variable	Default	Description
OZAKI	2	0=bypass (BLAS), 1=mantissa slicing, 2=CRT, 3=adaptive
OZAKI_COMPLEX	(auto)	Complex dispatch: 0=BLAS, 1=CPU, 2=GPU+fallback. Auto: 2 if on
OZAKI_N	(auto)	Slices (Sch.1: fp64=8, fp32=4) or primes (Sch.2: fp64=16, fp32=9)

OZAKI=3 (adaptive) starts with Scheme 1 on the first GPU call to learn the effective cutoff from preprocessing occupancy. Subsequent calls compare the Scheme-1 pair count (at the cached cutoff) against the Scheme-2 prime count and pick the cheaper path. On CPU, adaptive falls back to Scheme 2.

Accuracy

Variable	Default	Description
OZAKI_FLAGS	3	Sch.1 bitmask: 1=Triangular, 2=Symmetrize, 0=full S^2 square
OZAKI_TRIM	0	Levels to trim (0=exact). ~7 bits/level (Sch.1), ~4 bits (Sch.2)
OZAKI_I8	0	Sch.2: signed i8 residues (moduli<=128) instead of u8
OZAKI_GROUPS	0	Sch.2: K-grouping (0/1=off). Consecutive K panels, one reconstr.
OZAKI_MAXK	32768	Max K per preprocessing pass (0=full K in one pass)
OZAKI_THRESHOLD	12	Intensity threshold. Bypass when flops/(bytes*thr)<1. 0=always

GPU Path (requires LIBXSTREAM)

Variable	Default	Description
OZAKI_OCL	0	Enable OpenCL/GPU path (0=off, 1=on)
OZAKI_TM	(auto)	GPU output tile height (multiple of 8)
OZAKI_TN	(auto)	GPU output tile width (multiple of 16)

GPU-specific kernel tuning variables (OZAKI_RTM, OZAKI_RTN, OZAKI_WG, OZAKI_SG, OZAKI_KU, OZAKI_RC, OZAKI_PB, OZAKI_HIER, OZAKI_PREFETCH, OZAKI_SCALAR_ACC, OZAKI_DEVPOOL, OZAKI_CACHE) are documented in the LIBXSTREAM Ozaki README.

Monitoring and Diagnostics

Variable	Default	Description
OZAKI_VERBOSE	0	0=silent, 1=stats at exit, N=print every Nth GEMM call
OZAKI_STAT	0	Track C-matrix (0), A-representation (1), or B-representation (2)
OZAKI_DUMP	0	Dump A/B as MHD files at the given call-count (0=off)
OZAKI_EPS	inf	Dump A/B when epsilon error exceeds threshold
OZAKI_RSQ	0	Dump A/B when RSQ drops below threshold (updated after dump)
OZAKI_EXIT	1	Exit on accuracy violation after dump. 0=continue
OZAKI_PROFILE	0	Profile: 0=off, 1=all, 2=kernel, 3=preprocess A, 4=preprocess B

Benchmark

Variable	Default	Description
CHECK	0	Validate vs BLAS: 0=off, negative=auto-threshold, positive=custom
NREPEAT	3	Number of GEMM calls (first call is warmup when >1)
EVIL	0	Adversarial exponent-span test (see below)
OZAKI_DECAY	0	Decay diagnostic + K-permutation (0-4, see below)

Adversarial Input (EVIL) The EVIL variable initializes A and B with controlled exponent structure for stress-testing accuracy and adaptive slice reduction. The magnitude sets the exponent span in bits; the sign selects the distribution:

EVIL=N (N>0) Per-column. Column j of A is scaled by 2^(N*j/(ncols-1)), column j of B by the inverse. Product A*B is well-conditioned. Uniform exponents within each column.

EVIL=-N (N>0) Per-element. Each element gets a pseudorandom exponent in [0,N] via coprime shuffle, with opposite sign for B. Every row of A spans the full exponent range -- worst case for row-wise alignment and adaptive cutoff.

EVIL=0 Default shuffle mode (no exponent structure).

The per-column mode (EVIL>0) matches the NVIDIA emulation grading test (diagonal scaling with D and D^-1). The per-element mode (EVIL<0) is adversarial for the adaptive slice-pair reduction: it forces all slices to be populated in every row.

Decay Diagnostic and K-Permutation (OZAKI_DECAY) Controls K-dimension row reordering for smoothness and the forward-difference decay diagnostic. Values map to libxs_sort_t:

Value	Behavior
0	Off (default). No permutation, no diagnostic.
1	Identity permutation. Measure decay on original ordering.
2	Sort B rows by L1-norm, apply same K-permutation to A.
3	Sort B rows by mean value, apply same K-permutation to A.
4	Greedy nearest-neighbor chain on B rows (O(K^2*N)).

Values 2-4 compute a K-permutation from B (via libxs_sort_smooth) before Ozaki-1 slicing. Both A and B are sliced using the permuted K-order. Since C = A*B = (A*P^T)*(P*B) for any permutation matrix P, the result is mathematically identical -- only the int8 digit structure along K changes.

When nonzero, reports the forward-difference decay of int8 slice buffers along K, M, and N axes (first K-group only, single- threaded). Uses libxs_fprint per-axis mode internally. Output goes to stderr:

OZ1[MxNxK] Delta-K: d1=... d2=... ... OZ1[MxNxK] Delta-M: d1=... d2=... ... OZ1[MxNxK] Delta-N: d1=... d2=... ...

Decaying values indicate exploitable smoothness; growing values (~2x per order) indicate unstructured data where data-independent schemes (Ozaki-1, Ozaki-2) are well-matched.

Profiling

The OZAKI_PROFILE variable enables per-GEMM timing collected into a histogram reported at program exit:

```
OZAKI PROF: 850 DP-GFLOPS/s (17.0 INT8-TOPS/s, 20x)
```

Profile modes select which phase is measured:

Mode	CPU	GPU
1/negative	All phases (preprocess+dot)	All profiled kernels
2	Kernel only (dot products)	Dotprod kernel only
3	Preprocessing (A+B)	Preprocess A kernel
4	Preprocessing (A+B)	Preprocess B kernel

On the CPU, modes 3 and 4 are equivalent (A and B preprocessing is interleaved across OpenMP threads).

Example

```
./dgemm-wrap.x 256                          # CRT scheme (default)
OZAKI=1 ./dgemm-wrap.x 256                  # mantissa slicing
OZAKI_TRIM=4 OZAKI=1 ./dgemm-wrap.x 256     # drop 4 least significant diagonals
OZAKI=2 OZAKI_GROUPS=4 ./dgemm-wrap.x 4096  # CRT with K-grouping
OZAKI=3 ./dgemm-wrap.x 4096                 # adaptive scheme selection
EVIL=512 ./dgemm-wrap.x 1024                # accuracy grading (wide exponent span)
EVIL=1 OZAKI_PROFILE=1 ./dgemm-wrap.x 1024  # narrow span (shows pair savings)
EVIL=-52 ./dgemm-wrap.x 1024                # per-element span (worst for cutoff)
OZAKI_DECAY=1 OZAKI=1 ./dgemm-wrap.x 256    # decay diagnostic (no reorder)
OZAKI_DECAY=2 OZAKI=1 ./dgemm-wrap.x 256    # sort by norm + decay diagnostic
OZAKI_DECAY=4 OZAKI=1 ./dgemm-wrap.x 256    # greedy NN + decay diagnostic
```

Predict Samples

Four executables demonstrating fingerprint-guided prediction:

- **predict_params** -- Parameter prediction from structured CSV (GPU kernel tuning, configuration databases).
- **predict_sunspots** -- Timeseries forecasting via sliding-window kNN (monthly sunspot numbers, 1749-present).
- **predict_earthquakes** -- Spatial prediction of earthquake magnitude from location and depth (USGS catalog).
- **predict_discharge** -- River discharge forecasting via sliding-window kNN with day-of-year seasonality (USGS NWIS daily streamflow).

Build

```
make
```

Or from the LIBXS root:

```
make GNU=1 samples/predict
```

predict_params

Train a prediction model from a CSV file and save it for later use. Finds the optimal training fraction and polynomial order automatically. Reports validation quality on a held-out subset.

```
./predict_params.x [fraction] [auto|cat|interp] [-N] <csvfile> [modelfile]

fraction   Validation split 0..1 for quality report (default: 0.8).
           The full model always trains on all entries.
auto       Auto-detect mode per output (default).
cat        Force categorical (kNN) for all outputs.
interp     Force interpolation for all outputs.
-N         Max polynomial order for final build (default: 0 = auto).
csvfile    Delimited text file (semicolons, commas, or tabs).
           The first line may be a header (auto-skipped if non-numeric).
modelfile  Output path for the binary model.
           Default: derived from CSV basename (e.g., data.csv -> data.bin).
```

```
./predict_params.x ../../samples/smm/params/tune_multiply_PVC.csv
```

predict_sunspots

Timeseries forecasting using sliding-window nearest-neighbor prediction. The recent history (window of W values) serves as input; the next H values are predicted as output. The kNN confidence indicates whether similar patterns were seen in training.

```
./predict_sunspots.x <csvfile> [train_fraction]

csvfile         Semicolon-delimited timeseries (SILSO sunspot format).
train_fraction  Fraction of data used for training (default: 0.8).
```

```
./predict_sunspots.x predict_sunspots.csv 0.8

Loaded 3328 monthly sunspot values from predict_sunspots.csv
Window=12, Horizon=6, Train=2650, Test=666
Built: 51 clusters, 32.0x compression, order=2
Forecast quality (661 test windows):
  step   avg-err   max-err
  t+1      17.58     88.10
  t+2      19.48    115.00
  t+3      21.01    107.10
  t+4      21.76    114.50
  t+5      22.84    118.40
  t+6      24.26    153.20
  avg confidence: 1.000
```

Data Source Monthly mean total sunspot number from SILSO (World Data Center, Royal Observatory of Belgium). Semicolon-delimited: year, month, decimal_year, sunspot_number, std_dev, obs_count, marker. "Source: WDC-SILSO, Royal Observatory of Belgium, Brussels"

predict_earthquakes

Predict earthquake magnitude from geographic location and depth. This is a spatial prediction problem (not timeseries): given where an earthquake occurs, what magnitude is expected based on historical patterns at nearby locations?

```
./predict_earthquakes.x <usgs_csv> [train_fraction]

usgs_csv        USGS earthquake catalog CSV (comma-delimited).
train_fraction  Fraction of data for training (default: 0.8).
```

```
./predict_earthquakes.x predict_earthquakes.csv

Loaded 19619 earthquake events from predict_earthquakes.csv
Inputs: latitude, longitude, depth -> Output: magnitude
Train=15695, Test=3924
Built: 125 clusters, 83.9x compression, order=2
Prediction quality (3924 test events):
  avg magnitude error: 0.272
  max magnitude error: 2.700
  avg confidence: 0.649
```

Data Source USGS Earthquake Hazards Program (public domain, US Government). Comma-delimited: time, latitude, longitude, depth, mag, magType, ...

predict_discharge

River discharge (streamflow) forecasting using sliding-window kNN with day-of-year as an additional input dimension to capture seasonality. Uses log-transform on outputs (via API) for heavy-tailed data. Predicts the next 7 days from the previous 14 days + rate-of-change derivatives.

```
./predict_discharge.x <discharge_tsv> [train_fraction]

discharge_tsv   USGS NWIS daily discharge (tab-delimited RDB format).
train_fraction  Fraction of data for training (default: 0.8).
```

```
./predict_discharge.x predict_discharge.tsv

Loaded 9135 daily discharge values from predict_discharge.tsv
Window=14 (+3 diffs +day-of-year), Horizon=7, Train=7294, Test=1827
Built: 85 clusters, 58.4x compression, order=2
Forecast quality (1821 test windows):
  step   avg-err   max-err
  t+1      644.5   17726.6
  t+2      769.0   21363.5
  t+3      877.9   23199.8
  t+4      963.7   24193.4
  t+5     1044.1   25086.3
  t+6     1131.3   25735.3
  t+7     1225.9   26729.3
  avg confidence: 1.000
```

Data Source USGS National Water Information System (public domain, US Government). Colorado River at Lees Ferry, site 09380000. Tab-delimited RDB, comment lines start with #, data columns: agency_cd, site_no, datetime, discharge_value, qualification_code.

How It Works

All samples share the same prediction library (libxs_predict):

1. **predict_params**: Each CSV row is an independent (inputs, outputs) pair. The model learns spatial relationships in the input space and predicts outputs for unseen input combinations.
2. **predict_sunspots**: Uses `libxs_predict_set_series` to declare timeseries structure; the framework constructs sliding windows internally from accumulated timesteps. The model finds historically similar windows and predicts the continuation. The kNN confidence reflects how well the current pattern matches training history.
3. **predict_earthquakes**: Each earthquake event provides (lat, lon, depth) as inputs and magnitude as output. The model finds geographically similar past events and predicts expected magnitude for new locations.
4. **predict_discharge**: Combines temporal sliding-window (14 days) with day-of-year seasonality as an extra input dimension. Log-transform on outputs (via `libxs_predict_set_transform`) handles heavy-tailed discharge data transparently.

The fingerprint automatically determines per-output whether polynomial interpolation or distance-weighted kNN voting is more appropriate. Per-output confidence and variance scores enable the caller to gate predictions and fall back to safe defaults when the model is uncertain.

When confidence is low (<0.7), the framework automatically expands to multi-cluster blending, improving predictions by averaging over distinct regimes (e.g., -4% MAE on earthquake magnitude prediction).

Timeseries samples use `LIBXS_PREDICT_EXTRAPOLATE` mode which enables recency weighting (recent neighbors preferred), continuous-valued output without snap-to-nearest discretization, and local coherence smoothing across horizon steps.

The sunspot sample uses `libxs_predict_set_series` to declare timeseries structure: instead of manually constructing sliding windows, the caller pushes one timestep at a time (with outputs=NULL) and the framework builds all valid windows at build time. For multiple co-observed series, `libxs_predict_set_decompose(LIBXS_PREDICT_SPREAD)` transforms the stacked windows into sum/diff modes, exploiting anti-correlation between series (e.g., one's gain is the other's loss).

The sunspot sample additionally demonstrates forward-inverse-forward iteration: predicting outputs, finding the canonical historical window via `libxs_predict_inverse`, then re-predicting from that pattern. This reduces worst-case errors (-17% max error) at the cost of slight average regression -- a variance-bias tradeoff useful for applications where avoiding catastrophic predictions matters most.

Registry Microbenchmark

Benchmarks the performance of the LIBXS registry (key-value store) dispatch path under different access patterns and concurrency levels.

Programs

```
./registry.x [total [nrepeat [nthreads]]]
```

Argument	Default	Description
total	10000	Number of unique keys to register (min 2)
nrepeat	10	Repeat iterations for lookup phases
nthreads	max available	Number of OpenMP threads

Measurements:

- Duration to register (insert) all keys into the registry.
- Cold lookup with shuffled access pattern (defeats thread-local cache).
- Cached lookup with sequential/repeating pattern (hits thread-local cache).
- Multi-threaded parallel reads across all threads.
- Contended parallel writes (each thread registers its own key range).
- Mixed read/write: one writer thread while remaining threads read.

The multi-threaded benchmarks require at least 2 threads and are skipped when running single-threaded.

```
./registryf.x
```

Fortran variant with hardcoded parameters (10000 keys, 10 repeats). Measures registration, cold lookup, and cached lookup.

Scaling Behavior

Read-only accesses stay roughly constant in per-op duration due to the thread-local cache. Write accesses are serialized and duration scales with the number of threads.

Rosetta

Demonstrates hierarchical type discovery on opaque binary data. A table of harmonic-series values is encoded as a flat byte blob with no metadata. The analysis rediscovers the element type, record stride, and per-field structure -- recovering mathematical content from anonymous bytes.

What It Shows

The sample proceeds through the levels described in the algebra paper (Section "Hierarchical Type Discovery"):

Step	Tool used	What it finds
Flat probe	libxs_fprint (UNKNOWN)	inconclusive (expected)
Stride sweep	libxs_fprint per-column	f64, stride=6
Shuffle test	libxs_shuffle + fprint	order matters (3-4x)
Field analysis	libxs_fprint per-field	decay rates per column
Sort test	libxs_sort_smooth (GREEDY)	already optimally ordered
Verification	libxs_setdiff_min	0 unmatched, tol=0

Key observations from the output:

- The flat 1-D probe fails because interleaved fields of different scales (1/n mixed with n mixed with ln(n)) look like noise when read sequentially. This motivates the stride sweep.
- The stride sweep requires ALL columns at a candidate width to have decay < 1 before accepting it, which eliminates false positives from partial correlations at wrong strides.
- Shuffle stability confirms that the record order carries genuine structure (decay increases ~4x after coprime permutation).
- Per-field analysis reveals:
  - `[0]` decay=0 -- perfect ramp (sequential index)
  - `[1]` decay~0.63 -- 1/n (decaying but not ultra-smooth)
  - `[2]` decay~0.20 -- H_n (partial sums, very smooth)
  - `[5]` decay~0.29 -- converges to Euler-Mascheroni gamma
- GREEDY sort confirms the data is already optimally ordered (the natural 1..64 sequence is the smoothest permutation).

The Data

For n = 1, 2, ..., 64 the table stores six f64 fields per record:

```
[0] n            integer index
[1] 1/n          harmonic term
[2] H_n          partial sum of the harmonic series
[3] ln(n)        natural logarithm
[4] H_n - ln(n)  difference (converges to gamma from above)
[5] H_n - ln(n) - 1/(2n)   better gamma estimate
```

Total: 64 records x 6 fields = 384 doubles = 3072 bytes.

Build

```
cd samples/rosetta
make GNU=1
```

Usage

```
./rosetta.x
```

No arguments. The sample is self-contained: it generates the data, encodes it as a flat blob, runs the full analysis, and prints the results.

Example Output

```
ROSETTA: Recovering structure from opaque bytes
Ground truth (hidden from the analysis):
  64 records x 6 fields = 384 doubles (3072 bytes)
  Data: harmonic series H_n and derived quantities.
  Field [5] converges to Euler-Mascheroni gamma.
Level 0: Flat 1-D probe (all bytes as one sequence)
  Input: 3072 opaque bytes
  No type has decay < 1 -- flat stream is not smooth.
  This is expected: interleaved fields of different
  scales look like noise when read sequentially.
Stride sweep: discovering record layout
  Best type:   f64
  Best stride: 6 elements (48 bytes per record)
  Records:     64
  Avg decay:   0.288244
Shuffle stability (record-level)
  Avg decay (original): 0.288244
  Avg decay (shuffled): 1.103839
  Ratio:                3.8x
  -> Record order carries structure.
Per-field decay analysis
  [0] n           decay=0.000000
  [1] 1/n         decay=0.631512
  [2] H_n         decay=0.203226
  [3] ln(n)       decay=0.259621
  [4] H_n-ln(n)   decay=0.342853
  [5] gamma_est   decay=0.292251  (last=0.5771953203, gamma=0.5772156649)
  Smoothest: [0] n (perfect ramp, decay=0)
  -> This reveals a sequential index column.
  Most interesting: [5] gamma_est -- converges to a constant.
GREEDY sort test (64 rows x 6 cols)
  Data is already optimally ordered for row smoothness.
Verification: setdiff(original, blob)
  Unmatched: 0, tolerance: 0.00e+00
  -> Byte-perfect recovery.
```

Why This Is Interesting

From 3072 anonymous bytes the framework discovers:

1. The element type (f64) -- not i32, not f32, not i64.
2. The record structure (6-field records of 48 bytes each).
3. A sequential index column (field [0], decay = 0).
4. A column converging to Euler-Mascheroni gamma (field [5]).

5. That the natural ordering is already optimal (no resorting).

No metadata, no format knowledge, no human guidance. The hierarchical composition -- stride sweep with all-columns-must-pass filtering, shuffle stability, per-field decay ranking -- is what makes this possible. Any single tool alone would either fail (flat probe) or produce ambiguous results (stride sweep without the all-columns requirement accepts wrong strides).

Memory Allocation (Microbenchmark)

Benchmarks pooled scratch memory allocation (`libxs_malloc` / `libxs_free`) against the standard C library `malloc` / `free`. The pool-based allocator recycles memory after an initial warm-up, avoiding repeated system calls in steady state. A Fortran variant compares `libxs_malloc` against Fortran ALLOCATE / DEALLOCATE.

Building

```
cd samples/scratch
make GNU=1
```

Produces `scratch.x` (C) and, if a Fortran compiler is found, `scratchf.x`.

Usage (C)

```
./scratch.x [ncycles [max_nactive [nthreads]]]
```

Argument	Default	Description
ncycles	100	Number of allocation/deallocation cycles
max_nactive	4	Max concurrent live allocations per cycle (max 24)
nthreads	1	OpenMP threads (clamped to omp_get_max_threads)

Environment Variables

Variable	Default	Description
CHECK	0	Non-zero: memset each allocation (validates writeable)

Compile-Time Knobs

Macro	Default	Description
MAX_MALLOC_MB	100	Upper bound on individual allocation size in MB
MAX_MALLOC_N	24	Array size for random-number table and alloc slots

```
## default: 100 cycles, up to 4 active allocations, 1 thread
./scratch.x

## heavy load: 500 cycles, up to 12 active allocations, 4 threads
./scratch.x 500 12 4

## validate pages are writable
CHECK=1 ./scratch.x 43 8 4
```

Usage (Fortran)

```
./scratchf.x [mbytes [nrepeat]]
```

Argument	Default	Description
mbytes	100	Allocation size in MB
nrepeat	20	Number of repetitions

Single-threaded benchmark comparing `libxs_malloc` / `libxs_free` against Fortran ALLOCATE / DEALLOCATE. Reports per-iteration times (ms) and an average speedup ratio.

What Is Measured

Each cycle draws a random number of allocations (1 to max_nactive) with random sizes (1 to MAX_MALLOC_MB MB each). The cycle allocates all buffers, optionally touches them (CHECK), then frees them.

Both paths use the same randomized sequence. An untimed warm-up cycle runs before the measured loops. Reported metrics:

- calls/s (kHz) -- allocation+free throughput for each allocator
- Scratch size -- high-water mark of the pool (MB)
- Malloc size -- peak aggregate allocation per cycle (MB)
- Scratch Speedup -- ratio of pool throughput to stdlib throughput
- Fair -- size-adjusted speedup: (malloc_size / scratch_size) * speedup

Shuffle

Benchmarks three shuffling strategies and compares their throughput:

Label	Method
RNG-shuffle	Fisher-Yates via `libxs_rng_u32` (reference baseline)
DS1-shuffle	`libxs_shuffle` -- deterministic in-place (coprime stride)
DS2-shuffle	`libxs_shuffle2` -- deterministic out-of-place (src to dst)

Each iteration works on an array of unsigned integers initialized to 0..n-1. On the final iteration the shuffled result can optionally be written as an MHD image file or analyzed for randomness quality.

Build

```
cd samples/shuffle
make GNU=1
```

Usage

```
./shuffle.x [nelems [elemsize [niters [repeat]]]]
```

Positional	Default	Description
nelems	64 MB / elemsize	Number of elements
elemsize	4	Element size in bytes (1, 2, 4, or 8)
niters	1	Shuffle passes per timed measurement
repeat	3	Timed iterations (first is warmup)

Environment Variables

Variable	Default	Description
RANDOM	0	Non-zero: count inversions and report randomness (rand=%)
SPLIT	1	Partitioning depth for the STATS metrics
STATS	0	Non-zero: print partition imbalance (imb) and distance (dst)

```
## Default run (64 MB of 4-byte elements, 3 repeats)
./shuffle.x

## 1M elements of 8 bytes, 4 shuffle passes, 5 repeats
./shuffle.x 1000000 8 4 5

## Enable randomness quality metric
RANDOM=1 ./shuffle.x 10000 4 1 3

## Enable partition statistics with split depth 2
STATS=1 SPLIT=2 ./shuffle.x
```

What Is Measured

Reported bandwidth is 2 * data-size / time (in MB/s). The factor of two accounts for reading and writing each element during the shuffle. The first iteration is excluded from the average.

Optional quality metrics:

- rand -- Enabled by RANDOM=1. Inversions are counted via merge-sort and compared against the expected count for a random permutation (n*(n-1)/4) to give a percentage.
- dst -- Manhattan distance of the element sum from the expected uniform value, split into hierarchical partitions (SPLIT).
- imb -- Partition imbalance, measuring how unevenly element sums are distributed across sub-partitions (SPLIT).

MHD Image Output When RANDOM is not set and the element size is 1, 2, 4, or 8 bytes, the final shuffled array is written as an MHD image per method (shuffle_rng.mhd, shuffle_ds1.mhd, shuffle_ds2.mhd). The images are shaped close to square (isqrt(n) x n/isqrt(n)) and converted to unsigned 8-bit via modulus.

Compile-Time Knobs

Variable	Default	Description
OMP	0	Enable OpenMP (not used by this sample)
SYM	1	Include debug symbols (-g)
BUBBLE_SORT	undefined	Define (-DBUBBLE_SORT) for O(n^2) inversion counter

Synchronization Primitives

Micro-benchmark for the lock implementations provided by LIBXS (`libxs_sync.h`). Measures single-thread latency (uncontended acquire/release) and multi-thread throughput (mixed read/write workload) for every compiled lock kind:

Lock kind	Description
LIBXS_LOCK_DEFAULT	Compile-time default (typically atomic)
LIBXS_LOCK_SPINLOCK	OS-native or CAS-based spin lock (if available)
LIBXS_LOCK_MUTEX	OS-native mutex / pthread_mutex_t (if available)
LIBXS_LOCK_RWLOCK	Reader/writer lock / pthread_rwlock_t

The default lock is always benchmarked. The remaining kinds are conditionally compiled depending on platform support.

Build

```
cd samples/sync
make GNU=1
```

OpenMP is enabled by default (OMP=1) for multi-threaded tests.

Run

```
./sync.x [nthreads] [wratio%] [work_r] [work_w] [nlat] [ntpt]
```

Argument	Default	Description
nthreads	all available	Number of OpenMP threads
wratio%	5	Percentage of write operations (0-100)
work_r	100	Simulated work inside read-critical section (cy)
work_w	10 * work_r	Simulated work inside write-critical section
nlat	2000000	Iterations for latency measurement
ntpt	10000	Iterations per thread for throughput measurement

```
./sync.x 4 5 100 1000

LIBXS: default lock-kind "atomic" (Other)

Latency and throughput of "atomic" (default) for nthreads=4 wratio=5% ...
        ro-latency: 11 ns (call/s 91 MHz, 33 cycles)
        rw-latency: 11 ns (call/s 90 MHz, 33 cycles)
        throughput: 0 us (call/s 9128 kHz, 328 cycles)
```

Measurement Details

- RO-latency: uncontended read-lock acquire/release pairs (4x unrolled), reported as nanoseconds per operation and TSC cycles.
- RW-latency: uncontended write-lock acquire/release pairs (4x unrolled).
- Throughput: all threads run a mixed read/write workload governed by wratio%. Simulated work inside the critical section is subtracted so only synchronization overhead is reported.

SYRK / SYR2K Sample

Demonstrates symmetric rank-k and rank-2k updates using the LIBXS dispatch-and-call model. The sample validates correctness against a plain Fortran reference and reports performance.

Operations

SYRK: C := alpha * A * A^{T} + beta * C (lower triangle) SYR2K: C := alpha * (A * B^{T} + B * A^T) + beta * C (upper triangle)

Only the specified triangle of C is written; the other triangle is left untouched.

Build

```
make
```

Requires a Fortran compiler (gfortran, ifort, or ifx). The Makefile picks up the compiler from the top-level Makefile.inc. MKL or BLAS linkage is enabled (BLAS=1) so that dispatch can use JIT-compiled kernels when available.

Run

```
./syrkf.x [N [K [nrepeat]]]
```

Arguments (all optional, positional):

```
N         Matrix dimension of C (N x N).  Default: 64
K         Inner dimension (columns of A).  Default: N
nrepeat   Number of timed repetitions.     Default: 100
```

Example Output

```
syrk(F): N=64 K=64 nrepeat=100

--- libxs_syrk (lower) ---
  max error (lower): 0.00000E+00

--- libxs_syr2k (upper) ---
  max error (upper): 0.00000E+00

--- SYRK performance ---
  time:      0.002 s (100 calls)
  perf:      28.4 GFLOPS/s
```

Notes

- The dispatch step (libxs_syrk_dispatch / libxs_syr2k_dispatch) returns a pointer to a registry-owned config. This pointer remains valid until libxs_finalize or the registry is destroyed. There is no need to release it manually.
- Internally, SYRK/SYR2K decompose into GEMM tiles on the diagonal and off-diagonal blocks. The dispatched GEMM kernel (MKL JIT, LIBXSMM, or fallback BLAS) handles the inner loop.
- Scratch memory for the temporary full-panel product is managed via a thread-local buffer that grows on demand and is freed at finalization.